Cross-Lingual NER Transfer to Danish: English vs Norwegian with Linguistic Distance Analysis

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Abstract

Named entity recognition (NER) is the task of locating and classifying real-world references in text, such as people, locations, and organizations, and is a core component of many downstream NLP applications. Building NER systems for low-resource languages is difficult because manual annotation is expensive and often unavailable. Cross-lingual transfer offers a practical alternative: train a model on a well-resourced language and deploy it on the target language without any target-side labels.

This paper investigates whether typological proximity improves zero-shot NER transfer to Danish. Specifically, we compare Norwegian and English as source languages under two multilingual encoders (mBERT and XLM-R), asking whether Norwegian's structural closeness to Danish translates into a measurable transfer advantage. All experiments are replicated across multiple random seeds to ensure reliability of the conclusions.[1]

1 Introduction

Named entity recognition (NER) identifies text spans referring to persons, locations, and organizations. For Danish, the bottleneck is data rather than model capacity: English offers abundant labeled resources; Danish does not (Hvingelby et al., 2020). Cross-lingual transfer addresses this by training on a well-resourced language and deploying to the target with no labels, or before any exist.

The key question is which source language best serves Danish. Norwegian is the natural candidate: both are North Germanic languages sharing substantial syntax, morphology, and vocabulary. English is the conventional high-resource default. We quantify this contrast using `lang2vec` WALS syntactic feature vectors (Littell et al., 2017): Norwegian is 5.7× closer to Danish than English (cosine distance 0.021 vs. 0.119), with Swedish intermediate at 0.039 and German more distant at 0.124. Swedish and German serve as reference points confirming the coherence of the North Germanic cluster and anchoring the broader Germanic comparison. If structural similarity shapes representation alignment inside multilingual encoders, Norwegian-trained models should transfer more effectively. Transfer from English to Danish has been shown to work (Plank, 2020); our contribution is a controlled Norwegian comparison with explicit typological grounding, holding encoder family, data format, and evaluation protocol constant.

We run five experiment families: monolingual source baselines, zero-shot transfer to Danish, transfer plus Danish fine-tuning, Danish-only supervision at varying data budgets, and a typological-distance analysis. Encoder family is fixed (mBERT and XLM-R) across all conditions so that performance differences are attributable to source language. A secondary aim is to disentangle three factors that are typically conflated: source-language choice, encoder architecture, and the amount of Danish supervision available. All comparisons span three random seeds with paired resampling tests.

Main research question. *Does lower typological distance to Danish predict better NER transfer: is Norwegian a more effective zero-shot source than English?*
RQ1. How do monolingual baselines compare to cross-lingual transfer routes?
RQ2. What is lost in zero-shot transfer, and how much does Danish fine-tuning recover?

2 Related Work

Cross-lingual NER is a crowded area, especially for low-resource targets. English-to-Danish transfer has been shown to work as a practical zero-shot baseline, improving quickly once even modest Dan-

[1] Code: https://github.com/Erikzo03/NLP-Project

ish supervision is added (Plank, 2020). Our results broadly corroborate that picture while adding a direct Norwegian comparison. Structural similarity often predicts transfer success (Fu et al., 2023), a claim we pressure-test with a controlled setup that holds architecture, data format, and evaluation protocol constant.

Two Danish-specific resources anchor the fully supervised baseline literature. DaNE (Hvingelby et al., 2020) provides the manually annotated corpus underlying most Danish NER work; DaCy (Enevoldsen et al., 2021) demonstrates strong Danish-supervised pipelines on the same conventions. Our Danish monolingual baselines are most directly comparable to that line. Crucially, the Universal NER benchmark (Mayhew et al., 2024) uses consistent label sets across all three languages, eliminating annotation-schema mismatches as a confound, a detail that strengthens cross-lingual comparisons.

Multilingual encoders make zero-shot transfer viable. XLM-R (Conneau et al., 2020) typically outperforms mBERT (Devlin et al., 2019) on cross-lingual tasks thanks to a larger and more diverse pretraining corpus; whether that edge persists when source and target are as close as Norwegian and Danish is an open question our design addresses.

Typological distance as a predictor of transfer has been explored via URIEL+ (Khan et al., 2025) and DistALS (Van Der Goot et al., 2025); recent work argues surface similarity explains much of the zero-shot gap (Sohn et al., 2024). We compute WALS-feature distances explicitly and tie them directly to observed F1 differences.

3 Dataset

We use the Universal NER benchmark (Mayhew et al., 2024) for English, Norwegian, and Danish, provided sentence-by-sentence in token-aligned IOB2 format with three entity types: PER (person), LOC (location), and ORG (organization).

Figure 1 illustrates the sentence-count imbalance across splits. English provides roughly five times as many training sentences as Danish or Norwegian, a data-volume advantage that is a separate factor from typological proximity, and one the fine-tuning experiments help disentangle. Danish and Norwegian are closely matched in size, which simplifies interpretation of the data-budget conditions.

The Danish data-budget conditions (10%, 25%, 50% of 4,383 training sentences) correspond to

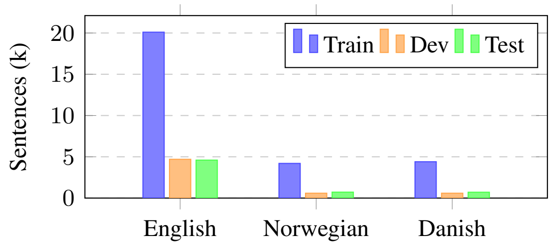

Figure 1: Universal NER sentence counts (thousands) per language and split. English provides $\sim 5\times$ more training data than Norwegian or Danish.

approximately 440, 1,100, and 2,200 sentences respectively, each a practically realistic annotation effort that tests how quickly target-language supervision closes the zero-shot gap. Because the primary annotation column uses the same label inventory across all three languages, observed performance differences are attributable to language transfer rather than annotation-schema mismatches.

4 Methodology

4.1 Experimental design

We run five families of experiments:

- monolingual baselines for English, Norwegian and Danish,
- zero-shot transfer to Danish from English and Norwegian,
- transfer fine-tuning, where a source-trained model is further trained on Danish,
- Danish-only supervision at multiple data budgets (10%, 25%, 50%, 100%), and
- linguistic-distance analysis based on WALS typological feature vectors.

The label space and Danish target splits stay fixed throughout. What varies are source language, encoder, and Danish supervision budget.

4.2 Models and checkpoints

Two multilingual encoders serve as the backbone for token classification:

- **mBERT**: `bert-base-multilingual-cased` (Devlin et al., 2019), 12 layers, 110M parameters, trained on 104 languages.
- **XLM-R**: `xlm-roberta-base` (Conneau et al., 2020), 12 layers, 125M parameters, larger and more diverse pretraining corpus.

Both encoders are fine-tuned for token classification using the Hugging Face Trainer API following standard cross-lingual transfer practice (KD-Nuggets, 2023). For fine-tuned transfer (EN→DA FT, NO→DA FT) we initialize from the best zero-shot checkpoint and continue training on Danish.

4.3 Training hyperparameters

All runs share the same configuration: learning rate 2×10^{-5} (AdamW), batch size 8 (train) / 16 (eval), max sequence length 256, 4 epochs, weight decay 0.01, early stopping by dev F1. We deliberately avoided per-language hyperparameter tuning; that choice may leave performance on the table for individual systems, but it reduces degrees of freedom in the source-language comparison (though per-language tuning could alter relative rankings). Multi-seed replication and its implications are discussed in Section 6.

4.4 Evaluation

The primary metric is strict span-level F1 (`span_f1.py`), requiring exact match on both entity boundaries and type. We also report unlabeled span F1 (exact boundaries, any label) and loose span F1 (partial overlap, matching label).

Subword-to-word label alignment follows the standard convention: the entity label is assigned to the first subword token of each word; subsequent subword tokens are masked in the loss, keeping token-level predictions consistent with word-level gold annotations.

Because the English test split is masked, test-set comparisons focus on Danish and Norwegian targets.

4.5 Statistical testing

We apply paired resampling tests (paired bootstrap, 10,000 iterations) to dev and test prediction files to assess the English–Norwegian contrast. With only a few seeds, these tests serve as a practical reliability check rather than a substitute for broader replication.

5 Results

5.1 Multi-seed zero-shot results

Table 1 reports the core comparison (Norwegian vs. English zero-shot transfer) across seeds 42, 1, and 2, with per-system means, standard deviations, and paired t-test p-values.

System	S42	S1	S2	Mean	Std	p
Development set (Danish dev)						
NO→DA XLM-R	86.49	84.98	84.90	85.45	0.89	0.632 ns
EN→DA XLM-R	85.52	86.57	85.52	85.87	0.60	
NO→DA mBERT	83.38	83.40	84.52	83.77	0.65	0.650 ns
EN→DA mBERT	—	82.23	83.68	82.95	1.02	
Test set (Danish test)						
NO→DA XLM-R	82.46	83.47	83.41	83.11	0.56	0.100 ns
EN→DA XLM-R	81.86	81.15	80.75	81.25	0.56	
NO→DA mBERT	82.88	84.06	80.96	82.63	1.56	**0.046***
EN→DA mBERT	—	78.98	79.56	79.27	0.41	

Table 1: Multi-seed strict F1 (%) for zero-shot transfer to Danish. "—" marks the EN mBERT seed-42 dev run, a known artifact excluded from comparisons (see Section 6.3). p-values from paired t-tests (NO vs EN within encoder). * $p < 0.05$.

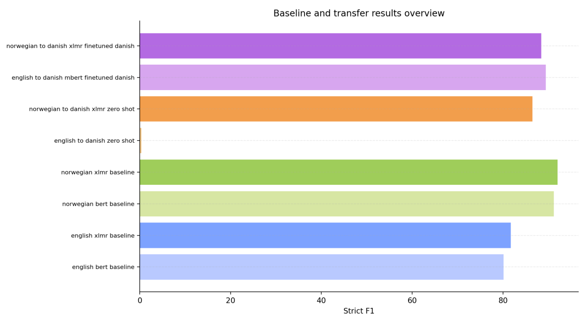

Figure 2: Overview of zero-shot and fine-tuned transfer results. Norwegian zero-shot models consistently outperform their English counterparts; fine-tuning narrows the gap substantially.

Seed 42 made Norwegian XLM-R look clearly better on dev (86.49 vs. 85.52), until averaging across seeds, at which point English XLM-R pulls slightly ahead (85.87 vs. 85.45). That reversal is the cautionary finding: single-seed development comparisons can mislead. On the held-out test set Norwegian leads for both encoders: +1.86 F1 for XLM-R ($p = 0.10$) and +3.36 F1 for mBERT ($p = 0.046$). The mBERT result is the more reliable: larger margin, lower variance, and conventional significance.

5.2 Transfer and fine-tuning results

Table 2 reports strict, unlabeled, and loose span F1 on held-out test sets. Full single-seed development results appear in Appendix Table 5. After Danish fine-tuning, Norwegian and English transfer models converge in a narrow band (≈84.7–86.6 strict F1), while zero-shot systems still trail by 2–4 points. Figure 2 provides a compact visual summary of all results.

Setup	Strict F1	Unlabeled F1	Loose F1
Source-language baselines			
Norwegian BERT	92.2	94.6	93.0
Norwegian XLM-R	91.4	94.4	92.9
Danish BERT	85.3	90.2	86.6
Danish XLM-R	84.9	89.9	87.2
Zero-shot transfer to Danish			
NO→DA XLM-R	82.5	88.7	84.0
NO→DA mBERT	82.9	88.8	84.4
EN→DA XLM-R	81.9	87.7	83.1
Fine-tuned on Danish			
NO→DA mBERT + DA FT	86.6	90.2	88.1
NO→DA XLM-R + DA FT	86.5	90.4	87.7
EN→DA mBERT + DA FT	85.5	89.5	87.0
EN→DA XLM-R + DA FT	84.7	88.2	87.1
Danish data efficiency			
Danish XLM-R 50%	83.6	88.9	85.6
Danish XLM-R 25%	84.1	88.6	86.7
Danish XLM-R 10%	72.8	78.2	84.4

Table 2: Test-set span F1 scores (%). Danish-target experiments evaluated on the Danish DDT test split; Norwegian baselines on the NorNE test split. English monolingual test results omitted (test split is masked).

System	PER	LOC	ORG
Danish BERT baseline	91.6	91.3	82.4
Danish XLM-R baseline	93.7	89.3	79.6
EN→DA XLM-R zero-shot	89.8	86.5	77.3
NO→DA XLM-R zero-shot	91.0	90.0	74.9
EN→DA mBERT + DA FT	91.0	92.7	83.5
NO→DA mBERT + DA FT	89.6	90.5	81.7
NO→DA XLM-R + DA FT	93.1	90.6	78.8

Table 3: Per-entity-type strict F1 (%) on Danish dev. MISC is excluded: those annotations reside in a secondary label column not read by training or evaluation code (see Section 6.3).

5.3 Per-entity-type F1

Table 3 shows per-type strict F1 for key systems on the Danish development set. ORG is the most variable category, swinging from 74.9 to 83.5 across systems, a spread far exceeding PER or LOC. The EN→DA mBERT + DA FT model records the highest ORG score (83.5), contributing to its strong overall ranking. LOC shows a notable zero-shot advantage for Norwegian XLM-R (90.0 vs. 86.5 for English), plausibly because place names share more lexical surface form between Norwegian and Danish. PER is the most stable category (89.6–93.7), consistent with person names transferring reliably across languages.

5.4 Typological distance and transfer gain

Table 4 places WALS distances alongside the corresponding zero-shot transfer scores. The ordering is directionally consistent: Norwegian's 5.7× proximity advantage (0.021 vs. 0.119) aligns with its

Source	WALS Distance	Zero-shot F1 (dev)	Test F1
Norwegian XLM-R	0.021	86.5	82.5
Norwegian mBERT	0.021	83.4	82.9
English XLM-R	0.119	85.5	81.9

Table 4: WALS syntactic distance vs. zero-shot transfer F1. Norwegian is 5.7× closer to Danish than English; the transfer advantage is directionally consistent and significant for mBERT (Table 1).

test-set lead for both encoders. The effect is modest in magnitude and falls short of significance for XLM-R, suggesting that structural proximity is a contributing factor rather than the sole determinant; encoder capacity and pretraining data volume also play a role.

6 Discussion

6.1 Multi-seed replication

Single-seed results mislead. Seed 42 made Norwegian XLM-R look clearly better on dev (86.49 vs. 85.52), but averaged across three seeds English XLM-R pulls slightly ahead (85.87 vs. 85.45). On the held-out test set the direction reverses in Norwegian's favor for both encoders. The takeaway: held-out test evaluation and multi-seed replication are both essential for source-language claims.

6.2 Typological distance in context

The zero-shot advantage of Norwegian aligns with its 5.7× lower WALS distance to Danish (Table 4), but two findings complicate a purely distance-based account. First, English mBERT fine-tuned on Danish achieves the best overall result (89.4 dev, 85.5 test), outperforming all Norwegian variants. English provides ∼5× more source-language training sentences; that initialization advantage outweighs typological proximity when target supervision is available to steer the model. The distance hypothesis is most informative in the pure zero-shot setting. Second, XLM-R generally outperforms mBERT on source baselines and zero-shot (e.g. 92.0 vs. 91.2 on Norwegian), reflecting its larger pretraining corpus, but the gap narrows after Danish fine-tuning. Unlabeled and loose F1 exceed strict by 3–5 points across systems, indicating that errors are mainly boundary or label near-misses rather than entirely missed spans. The largest gap at 10% Danish data (strict 70.0, loose 81.7) suggests that limited supervision lets the model detect spans but not reliably label them.

6.3 Pipeline notes

MISC F1 = 0.0. The Universal NER Danish file stores MISC annotations in a secondary column (e.g. `B-MISC#O`) not read by training or evaluation code, a data-format artifact rather than a modeling failure (32 dev and 44 test instances affected).

Anomalous EN→DA run. One seed-42 English-to-Danish dev prediction file contains 2,001 sentences against 564 in the gold (strict F1 ≈0.3% vs. 79.3 logged during training), caused by an incorrect dev-evaluation path that pointed at the English development split. A blast-radius check over all 17 prediction files found the mismatch isolated to this single file; all other runs are consistent (see Appendix Figure 4). This run is excluded from seed-42 dev comparisons; its test prediction is unaffected (78.8 test F1).

6.4 Future directions

Figure 3 shows that roughly 25–50% of the Danish training set suffices to match or exceed the zero-shot baselines, suggesting a practical strategy: start with a Norwegian zero-shot model, then annotate a modest fraction of the target domain. The most valuable extension would be adding Swedish as a source language to test whether the WALS distance gradient (NO 0.021 < SV 0.039 < EN 0.119) produces a corresponding F1 gradient across more than two source points.

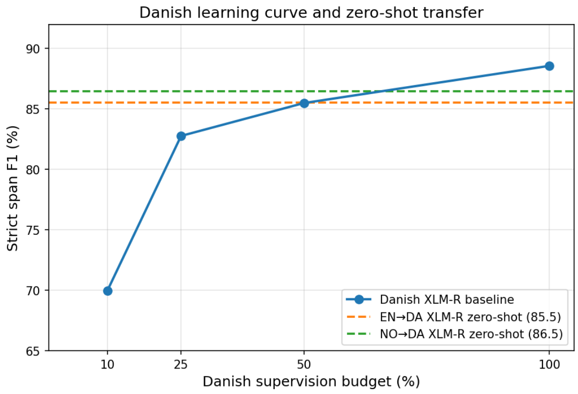

Figure 3: Danish XLM-R learning curve and zero-shot reference lines. Approximately 25–50% of Danish training data suffices to match the zero-shot baselines.

7 Conclusion

We ran a multi-seed study of cross-lingual NER transfer to Danish, comparing English and Norwegian as source languages under mBERT and XLM-R, evaluated with strict, unlabeled, and loose span F1 on development and held-out test sets.

Main finding. In the zero-shot setting, Norwegian is the better source on the Danish test set: +3.36 F1 for mBERT ($p = 0.046$, significant) and +1.86 for XLM-R ($p = 0.10$, directionally consistent). Norwegian is 5.7× closer to Danish in WALS syntactic-feature space (0.021 vs. 0.119), consistent with its zero-shot advantage.

When English outperforms. Once Danish fine-tuning is available, English mBERT achieves the best result (89.4 dev, 85.5 test): its ∼5× larger source corpus outweighs typological proximity given sufficient target supervision. The distance hypothesis is most informative in the pure zero-shot setting.

Practical recommendation. For zero-shot Danish NER, Norwegian is the empirically and linguistically motivated choice. If any Danish annotation is available, fine-tuning English mBERT on Danish is a strong practical baseline. Full Danish supervision remains the performance ceiling.

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A Development-Set Results

Table 5 reports single-seed (seed 42) development span F1 for all evaluated systems. The anomalous EN→DA mBERT zero-shot run is excluded (see Section 6.3).

B Evaluation Consistency Check

Figure 4 compares strict F1 from saved prediction files with training-time evaluation F1 for all 17 runs. All runs except the known EN→DA artifact cluster tightly near the diagonal, confirming that the pipeline produced internally consistent results.

Setup	Strict F1	Unlabeled F1	Loose F1
Source-language baselines			
Norwegian XLM-R	92.0	95.9	92.8
Norwegian BERT	91.2	95.1	92.1
Danish BERT	89.0	93.0	92.2
Danish XLM-R	88.6	92.6	91.8
English XLM-R	81.7	86.3	85.7
Zero-shot transfer to Danish			
NO→DA XLM-R	86.5	90.3	90.7
EN→DA XLM-R	85.5	90.7	88.5
NO→DA mBERT	83.4	88.2	86.6
Fine-tuned on Danish			
EN→DA mBERT + DA FT	89.4	93.4	91.7
NO→DA XLM-R + DA FT	88.4	92.7	91.5
NO→DA mBERT + DA FT	87.6	92.1	90.5
EN→DA XLM-R + DA FT	87.0	90.8	90.4
Danish data efficiency			
Danish XLM-R 50%	85.5	90.8	90.0
Danish XLM-R 25%	82.8	88.8	88.3
Danish XLM-R 10%	70.0	77.2	81.7

Table 5: Development-set span F1 scores (%). Strict = exact span and label; Unlabeled = exact span, any label; Loose = partial overlap with matching label.

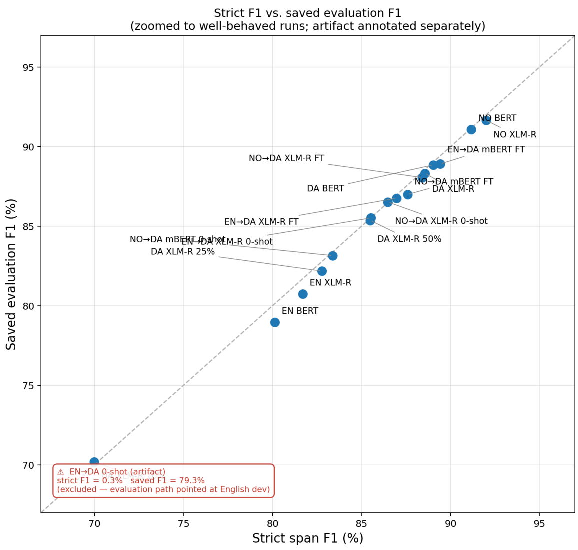

Figure 4: Strict F1 from saved predictions vs. training-time evaluation F1. The EN→DA zero-shot artifact (strict F1 ≈0.3%) is annotated but excluded from the axis range. All other runs fall within ∼1–2 points of the diagonal.

C Group Work and AI Usage

Group Contributions

All three group members contributed evenly to the project.

AI Tool Usage

Perplexity AI was used to identify potentially relevant papers for the Related Work section. All cited papers were subsequently verified and read independently by us. AI tools were used for LaTeX syntax and formatting only (e.g. table layout, figure placement). The scientific content, analysis, and conclusions are entirely our own.